

Turning Interference into Power: Intelligent Anti-Jamming for Ambient Backscatter via Imitation-Augmented MADDPG

Thai-Duong Nguyen¹, Thanh-Dao Nguyen¹, Thi-Thu Hoang², Viet-Hoang Vu¹,
and Ngoc-Tan Nguyen^{1*}

¹ VNU University of Engineering and Technology (VNU-UET), Hanoi, Vietnam

² Posts and Telecommunications Institute of Technology, Hanoi, Vietnam

Abstract. Ensuring robust and energy-efficient connectivity for IoT networks under malicious interference remains a critical challenge. While Ambient Backscatter (AmB) offers a low-power solution, coordinating multiple agents to mitigate jamming is non-trivial. This paper proposes a paradigm shift by exploiting high-power jamming signals not as threats, but as primary excitation sources for AmB transmissions. To solve the resulting decentralized coordination problem, we introduce the **Imitation-Augmented Multi-Agent Deep Deterministic Policy Gradient (IA-MADDPG)** framework. Unlike conventional MARL approaches that suffer from exploration inefficiency in high-dimensional continuous action spaces, our framework accelerates convergence by bootstrapping the policy via expert-guided behavior cloning. The agents learn a joint policy to dynamically optimize the reflection coefficient and communication mode (D2D/Relay). Simulation results demonstrate that IA-MADDPG outperforms conventional baselines, achieving a 32.8% increase in network reward and a near-perfect (98%) transmission success rate under intense jamming.

Keywords: Ambient Backscatter · Anti-Jamming · Multi-Agent DRL · Imitation Learning.

1 Introduction

Massive Internet of Things (IoT) deployments in 6G networks face a dual challenge: stringent energy constraints and vulnerability to malicious Radio-Frequency (RF) jamming [1]. Traditional anti-jamming techniques, such as Frequency Hopping, require active power consumption that is often prohibitive for battery-limited devices [2]. Ambient Backscatter Communication (AmB) offers a sustainable alternative by modulating existing RF waves [3]. In this context, we explore a counter-intuitive defense strategy: rather than evading high-power jamming signals, we exploit them as primary excitation sources to energize backscatter transmissions, effectively repurposing the adversary’s energy for communication [4].

* Corresponding author

However, orchestrating this system poses a non-trivial decentralized control problem. Agents must operate under partial observability while optimizing a complex trade-off: dynamically selecting between Device-to-Device (D2D) and Relay modes, and fine-tuning the continuous reflection coefficient (α) to balance signal strength against circuit energy costs. Static heuristics and greedy strategies typically fail to capture the coupled interference dynamics inherent in such multi-agent environments [5].

Multi-Agent Deep Reinforcement Learning (MADRL), specifically Multi-Agent Deep Deterministic Policy Gradient (MADDPG) [6], provides a framework for continuous control in these settings. However, standard MADDPG suffers from poor sample efficiency and "cold-start" instability, where agents struggle to discover viable policies in the early stages. To address this, we propose the **Imitation-Augmented MADDPG (IA-MADDPG)** framework. Our approach mitigates exploration overhead by bootstrapping the training with an analytical expert policy and integrating Prioritized Experience Replay (PER) [7] to accelerate convergence.

The main contributions of this paper are:

- We formulate a jamming-resilient AmB network model where interference is strategically harvested to sustain hybrid D2D and Relay communications.
- We propose the IA-MADDPG framework, which incorporates expert-guided warm-up and behavior cloning to prevent convergence to suboptimal local equilibria.
- Simulation results demonstrate that IA-MADDPG achieves a 98% transmission success rate and a 32.8% increase in network reward compared to Direct Transmission and Frequency Hopping baselines.

2 System Model and Problem Formulation

2.1 Network Scenario

We consider a decentralized anti-jamming communication network utilizing ambient backscatter communication, as illustrated in Fig. 1. The network comprises a set of N secondary user pairs, denoted by $\mathcal{N} = \{1, \dots, N\}$, where each pair consists of a source user (S_i) and a destination user (D_i). Additionally, a relay base station (RBS) is deployed to facilitate cooperative transmission. The network operates in the presence of a malicious jammer that continuously broadcasts interference signals with power P_J to disrupt legitimate communications.

To align with energy-efficient constraints, the RBS operates in a semi-passive mode, harvesting energy from the ambient jamming signal to modulate and forward data. Similarly, each source node S_i leverages the jamming signal to backscatter data via two distinct transmission modes, denoted by $m_i \in \{0, 1\}$:

- **Mode 0 (Direct D2D):** S_i backscatters the signal directly to D_i .
- **Mode 1 (Relay-assisted):** S_i backscatters to the RBS, which then forwards the signal to D_i .

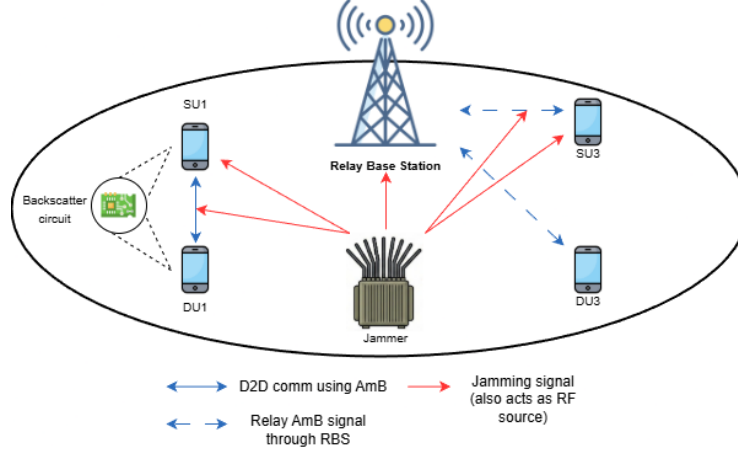


Fig. 1. System model: An Ambient Backscatter communication network operating under jamming interference.

2.2 Channel Model and SINR Analysis

Let $h_{X,Y} \in \mathbb{C}$ denote the channel coefficient between node X and node Y , and $g_{X,Y} = |h_{X,Y}|^2$ represent the corresponding channel gain. The received signal at destination D_i is modeled as:

$$y_{D_i} = \sqrt{P_J} h_{J,S_i} h_{S_i,D_i} \alpha_i x_{S_i} + n_{D_i}, \quad (1)$$

where $\alpha_i \in [0, 1]$ is the reflection coefficient controlled by S_i , x_{S_i} is the transmitted symbol, and n_{D_i} represents the additive white Gaussian noise (AWGN) with power N_0 .

The Signal-to-Interference-plus-Noise Ratio (SINR) for the *Direct D2D* mode is given by:

$$\gamma_i^{\text{D2D}} = \frac{GP_J g_{J,S_i} g_{S_i,D_i} \alpha_i^2}{P_J g_{J,D_i} + N_0}, \quad (2)$$

where G denotes the composite backscatter gain, accounting for antenna properties and reflection efficiency. The denominator term $P_J g_{J,D_i}$ represents the direct jamming interference at the receiver.

For the *Relay-assisted* mode, the end-to-end SINR is determined by the bottleneck link of the two-hop transmission:

$$\gamma_i^{\text{Relay}} = \min \left(\frac{GP_J g_{J,S_i} g_{S_i,R} \alpha_i^2}{P_J g_{J,R} + N_0}, \frac{GP_J g_{J,R} g_{R,D_i} \alpha_i^2}{P_J g_{J,D_i} + N_0} \right). \quad (3)$$

The energy consumption for agent i is formulated as $E_i = E_{\text{sense}} + \kappa \alpha_i^2$, where E_{sense} captures the circuit power for spectrum sensing and κ is a hardware-specific coefficient related to the backscatter impedance switching.

2.3 POMDP Formulation

Since each agent only possesses local information regarding the channel and interference, we formulate the anti-jamming problem as a decentralized Partially Observable Markov Decision Process (POMDP), defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \Omega \rangle$:

- **Observation Space (Ω_i):** Each agent i observes a local state $\mathbf{o}_i$, that contains $\gamma_i^{(t-1)}, E_i^{(t-1)}, m_i^{(t-1)}$, comprising the previous SINR, energy consumption, and transmission mode.
- **Action Space ($\mathcal{A}_i$):** The action vector is $\mathbf{a}_i = \{\alpha_i, m_i\}$, where α_i is continuous and m_i is discrete.
- **Reward Function (r_i):** To accelerate convergence and avoid sparse reward issues, we design a composite reward function that incorporates throughput, reliability margin, and energy penalty.

$$r_i(t) = w_1 \mathcal{R}_i(t) + w_2 \tanh\left(\frac{\gamma_i(t)}{\Gamma_{th}} - 1\right) - w_3 \alpha_i(t)^2 \quad (4)$$

Here, $\mathcal{R}_i(t) = \log_2(1 + \gamma_i(t))$ denotes the instantaneous spectral efficiency (throughput), where $\gamma_i(t)$ is the received SINR and the second term employs a hyperbolic tangent function, $\tanh(\cdot)$, acting as a soft QoS constraint. It provides a smooth gradient that guides the agent to satisfy the SINR threshold Γ_{th} . Finally, the third term, $-w_3 \alpha_i(t)^2$, introduces an energy penalty proportional to the reflection power, encouraging the agent to minimize circuit energy consumption by selecting the lowest sufficient reflection coefficient

3 Proposed IA-MADDPG Framework

Standard Multi-Agent Deep Deterministic Policy Gradient (MADDPG) often suffers from slow convergence and instability in high-dimensional continuous spaces, particularly due to the "cold-start" problem where agents explore randomly in the early stages. To address this, we propose the Imitation-Augmented MADDPG (IA-MADDPG) framework, which integrates expert knowledge injection with advanced experience replay mechanisms.

3.1 Imitation-Augmented Training Mechanism

The core innovation of IA-MADDPG is the integration of an analytical expert policy, denoted as μ^E , which greedily maximizes instantaneous SINR based on channel state information. This expert knowledge is leveraged in two phases:

Phase 1: Expert Warm-up Before the training process begins, we populate the Replay Buffer $\mathcal{D}$ with a set of high-quality transitions generated by μ^E . This creates a "demonstration dataset" that allows the agents to learn valid policies before interacting with the environment, significantly reducing the initial exploration of suboptimal actions.

Phase 2: Behavior Cloning (BC) Regularization During the training phase, we modify the actor’s objective function to not only maximize the critic’s Q-value but also minimize the divergence from the expert’s policy. The augmented loss function for the actor network μ_i of agent i is defined as:

$$\mathcal{L}(\theta_i^\mu) = \underbrace{-\mathbb{E}_{\mathbf{o}, \mathbf{a} \sim \mathcal{D}}[Q_i(\mathbf{o}, \mathbf{a}_1, \dots, \mu_i(\mathbf{o}_i), \dots, \mathbf{a}_N)]}_{\mathcal{L}_{\text{MADDPG}}} + \lambda_{\text{IL}} \underbrace{\mathbb{E}_{\mathbf{o} \sim \mathcal{D}}[\|\mu_i(\mathbf{o}_i) - \mu^E(\mathbf{o}_i)\|^2]}_{\mathcal{L}_{\text{BC}}}, \quad (5)$$

where $\mathcal{L}_{\text{MADDPG}}$ is the standard policy gradient loss, and $\mathcal{L}_{\text{BC}}$ is the Behavior Cloning loss (Mean Squared Error). The coefficient λ_{IL} is an annealing factor that decays over time, allowing the agent to initially mimic the expert and subsequently surpass it by exploring novel strategies via the RL component.

3.2 Prioritized Experience Replay and Smoothing

To further enhance stability, we employ *Prioritized Experience Replay (PER)*, which samples transitions with high Temporal Difference (TD) errors more frequently. Additionally, we incorporate *Target Policy Smoothing* (inspired by TD3) by adding clipped noise to the target actions during the critic update. This prevents the overfitting of Q-values to sharp peaks in the value landscape.

3.3 Algorithm Description

The complete training procedure for the proposed IA-MADDPG is summarized in Algorithm 1.

4 Performance Evaluation

4.1 Simulation Setup

The experimental environment is a $120 \times 120 \text{ m}^2$ area with $N = 5$ SU-DU pairs and a jammer with power $P_J = 1 \text{ W}$. The noise floor is $N_0 = 10^{-4} \text{ W}$. We utilize a standard backscatter path-loss model with exponent $\eta = 2.0$ and a reflection gain $G = 10^4$. The IA-MADDPG agents utilize an Actor-Critic architecture (Actor: [256, 256, 128], Critic: [512, 256, 128, 64]) trained over 450 episodes with a batch size of 256. We benchmark against three baselines:

1. **Direct Transmission (DT):** A static policy utilizing D2D mode with a fixed reflection coefficient $\alpha = 0.2$.
2. **Greedy Strategy:** A heuristic that dynamically selects the communication mode (D2D or Relay) to maximize instantaneous SINR.
3. **Frequency Hopping (FH):** A stochastic avoidance strategy with a success probability derived from the ratio of jammed to total channels.

Algorithm 1 Imitation-Augmented MADDPG (IA-MADDPG)

```

1: Input: Batch size  $B$ , Episodes  $M$ , Steps  $T$ , IL decay  $\lambda_{\text{decay}}$ .
2: Initialize: Actors  $\mu_i$ , Critics  $Q_i$ , Target networks  $\mu'_i, Q'_i$  for all  $N$  agents.
3: Initialize: Prioritized Replay Buffer  $\mathcal{D}$ , Expert Policy  $\mu^E$ .
4: // Phase 1: Expert Warm-up
5: Generate  $K$  transitions using  $\mu^E$  and store in  $\mathcal{D}$ .
6: // Phase 2: Interaction & Training
7: for episode = 1 to  $M$  do
8:   Reset environment, observe initial states  $\mathbf{o} = (\mathbf{o}_1, \dots, \mathbf{o}_N)$ .
9:   Dynamic Thresholding for Curriculum Learning
10:  for  $t = 1$  to  $T$  do
11:    Select actions  $\mathbf{a}_i = \mu_i(\mathbf{o}_i) + \mathcal{N}_t$  (Exploration noise).
12:    Execute actions, observe rewards  $\mathbf{r}$  and next states  $\mathbf{o}'$ .
13:    Store transition  $(\mathbf{o}, \mathbf{a}, \mathbf{r}, \mathbf{o}')$  in  $\mathcal{D}$  with max priority.
14:    Sample a minibatch of size  $B$  from  $\mathcal{D}$  using PER weights  $w_j$ .
15:    for each agent  $i = 1$  to  $N$  do
16:      1. Critic Update:
17:      Compute target actions with smoothing noise:  $\tilde{\mathbf{a}}'_k = \mu'_k(\mathbf{o}'_k) + \epsilon$ .
18:      Calculate target  $y_j = r_i^j + \gamma Q'_i(\mathbf{o}'^j, \tilde{\mathbf{a}}'_1^j, \dots, \tilde{\mathbf{a}}'_N^j)$ .
19:      Minimize Loss:  $L(\theta_i^Q) = \frac{1}{B} \sum_j w_j (y_j - Q_i(\mathbf{o}'^j, \mathbf{a}^j))^2$ .
20:      Update PER priorities based on TD-error.
21:      2. Actor Update:
22:      Compute gradient  $\nabla_{\theta_i^\mu} \mathcal{L}$  combining  $\mathcal{L}_{\text{MADDPG}}$  and  $\mathcal{L}_{\text{BC}}$  (Eq. 5).
23:      Update weights  $\theta_i^\mu$ .
24:    end for
25:    Update target networks:  $\theta' \leftarrow \tau\theta + (1 - \tau)\theta'$ .
26:     $\mathbf{o} \leftarrow \mathbf{o}'$ .
27:  end for
28:  Decay  $\lambda_{\text{IL}} \leftarrow \lambda_{\text{IL}} \cdot \lambda_{\text{decay}}$ .
29: end for

```

4.2 Numerical Results

Convergence Analysis and Learning Stability Fig. 2 (Top-Left) delineates the learning trajectory of the proposed IA-MADDPG framework compared to the deterministic baselines. A critical observation is the rapid convergence velocity of our approach. Despite the high-dimensional continuous action space, the agents leverage the imitation-augmented mechanism to escape suboptimal local equilibria effectively. The policy curve intersects the strongest baseline (Greedy Strategy) as early as episode 70, subsequently maintaining a robust upward trend before stabilizing at an asymptotic average reward of ≈ 1.95 . In contrast to the static behaviors of Direct Transmission (DT), Greedy, and Frequency Hopping (FH), which are capped at 1.3715, 1.4607, and 1.4551 respectively, IA-MADDPG achieves a substantial performance leap. Specifically, it yields a **33.4% improvement** over the Greedy strategy and a **42.1% gain** over DT. The stochastic oscillations observed in the later training stages (episodes 300-450) are indicative of the agents continuously exploring fine-grained adjustments

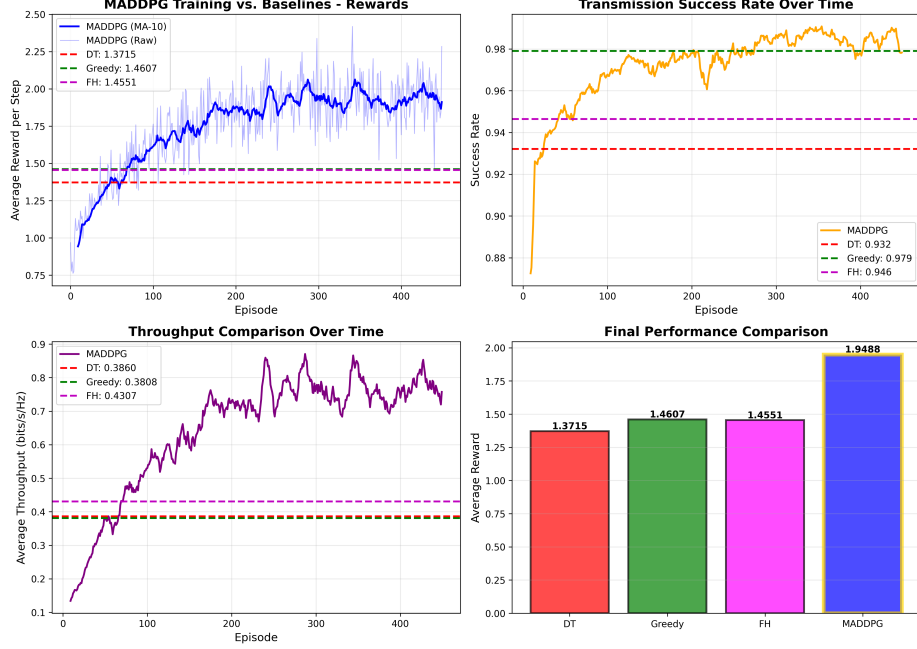


Fig. 2. Performance evaluation of the proposed IA-MADDPG against baseline strategies: (Top-Left) Training convergence of average reward; (Top-Right) Evolution of transmission success rate; (Bottom-Left) Average throughput analysis; (Bottom-Right) Final accumulated network reward comparison.

to the reflection coefficient α to maximize spectral efficiency without violating SINR constraints.

The Reliability-Throughput Trade-off The superiority of IA-MADDPG is most profound when analyzing the decoupling of reliability and throughput, as evidenced by Fig. 2 (Top-Right) and (Bottom-Left). Existing heuristics fail to optimize this trade-off simultaneously. As shown in Fig. 2 (Top-Right), the *Greedy* strategy achieves a commendable success rate of 0.979. However, a cross-examination with Fig. 2 (Bottom-Left) reveals that this reliability comes at the cost of spectral efficiency; Greedy yields a low average throughput of ≈ 0.38 bits/s/Hz, marginally lower than even the naive DT approach. This implies that the Greedy heuristic operates overly conservatively, selecting distinct modes or low reflection coefficients to ensure connectivity, thereby wasting harvestable interference energy. Conversely, the *Frequency Hopping* (FH) baseline improves throughput to 0.4307 bits/s/Hz but suffers in reliability (0.946) due to its inability to proactively sense and adapt to the jammer's dynamic presence. IA-MADDPG effectively navigates this Pareto frontier. By treating the jamming signal as a power source rather than purely noise, the agents learn to opportunistically maximize the backscatter rate during favorable channel conditions.

Consequently, our framework achieves a terminal throughput of approximately **0.75 bits/s/Hz**, nearly doubling the performance of the baselines, while sustaining a carrier-grade transmission success rate exceeding **0.98**. This confirms that the proposed method does not merely evade interference but intelligently reconfigures the network to turn the adversary’s high-power signals into a communication asset.

5 Conclusion

In this paper, we have proposed IA-MADDPG, a novel anti-jamming framework for Ambient Backscatter networks that strategically exploits jamming signals as a power source. By integrating Imitation Learning and Prioritized Experience Replay, the proposed approach addresses the challenges of slow convergence and instability in multi-agent environments. Simulation results demonstrate that our framework significantly outperforms conventional baselines, achieving a 32.8% increase in reward and a 98% transmission success rate. These findings confirm that intelligent, adaptive reconfiguration of backscatter parameters is essential for resilient IoT communications in hostile RF environments. Future work will investigate the impact of imperfect channel state information and extend the framework to large-scale, heterogeneous networks.

References

1. Tatara, Harsh, Shafi, Mansoor, Molisch, Andreas, Dohler, Mischa, Sjoland, Henrik, Tufvesson, Fredrik. (2021). 6G Wireless Systems: Vision, Requirements, Challenges, Insights, and Opportunities. Proceedings of the IEEE. PP. 1-34.
2. Pirayesh, Hossein, and Huacheng Zeng. "Jamming attacks and anti-jamming strategies in wireless networks: A comprehensive survey." IEEE communications surveys & tutorials 24, no. 2 (2022): 767-809.
3. N. Van Huynh, D. T. Hoang, X. Lu, D. Niyato, P. Wang, and D. I. Kim, "Ambient Backscatter Communications: A Contemporary Survey," IEEE Communications Surveys & Tutorials, vol. 20, no. 4, pp. 2889–2922, 2018.
4. N. Van Huynh, D. N. Nguyen, D. Thai Hoang, E. Dutkiewicz and M. Mueck, "Ambient Backscatter: A Novel Method to Defend Jamming Attacks for Wireless Networks" in IEEE Wireless Communications Letters, vol. 9, no. 2, pp. 175-178, Feb. 2020,
5. Jia, Luliang, Nan Qi, Zhe Su, Feihuang Chu, Shengliang Fang, Kai-Kit Wong, and Chan-Byoung Chae. "Game theory and reinforcement learning for anti-jamming defense in wireless communications: Current research, challenges, and solutions." IEEE Communications Surveys & Tutorials (2024).
6. Lowe, Ryan, Yi I. Wu, Aviv Tamar, Jean Harb, OpenAI Pieter Abbeel, and Igor Mordatch. "Multi-agent actor-critic for mixed cooperative-competitive environments." Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17). Curran Associates Inc., Red Hook, NY, USA, 6382–6393.
7. Schaul, Tom, John Quan, Ioannis Antonoglou, and David Silver. "Prioritized experience replay." arXiv preprint arXiv:1511.05952 (2015).